

$\Lambda_b \rightarrow \Lambda^0 \gamma$: v3 BDT review

Frozen 1,028-chunk Physics versus inclusive $Z \rightarrow b\bar{b}$ snapshot

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FCC-ee analysis review

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Validation scan and independent-test projection; score is a proposal for PI review.

Frozen inputs and candidate cutflow

Candidate stage	Direct signal	Zbb other
Stage 1	55,395	5,935,608
Fit check	55,395	4,843,426
Λ^0 check	55,264	4,401,877
Offline selected	54,872	1,060,544

- 100,000 signal input events; generator audit counts 100,002 direct decays.
- 376,723,929 processed Zbb events in 1,028 valid v3 chunks; 172 of 1,200 jobs missing at freeze.
- Offline: $|m(p\pi) - m_{\Lambda^0}| \leq 12.5$ MeV, $E(\Lambda_b) \geq 10.5$ GeV, same thrust hemisphere; no photon veto.

Normalization

$$S = 816642 (N_{\text{direct}} / N_{\text{gen events}})$$

$$B = 9 \times 10^{11} (N_{\text{Zbb other}} / N_{\text{processed Zbb}})$$

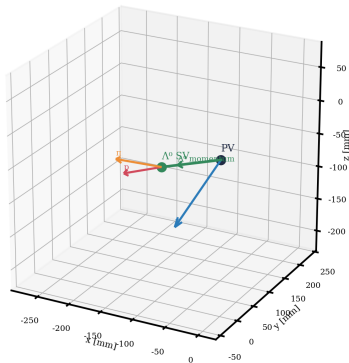
Candidate yields in the full 4.7–6.5 GeV mass interval. One forced decay/event is the code approximation; two input events contain an extra direct chain.

Full cutflow with conditional and cumulative retention: docs/STAGE2_V3_BDT_REVIEW_2026-10-03.md; frozen catalog and preparation summary in docs/data/stage2_v3_bdt_1028.

One reconstructed event: topology and isolation

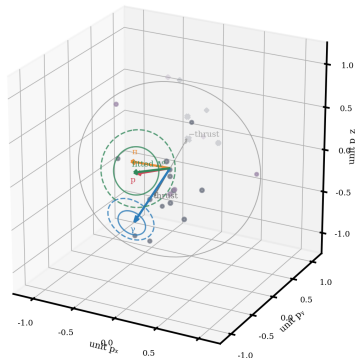
v3 reconstructed $\Lambda b \rightarrow \Lambda^0 \gamma$ candidate • tuple key 4438, raw entry 38885, slot 0
 $m(\Lambda \gamma) = 5.648$ GeV; same thrust hemisphere=True

Fitted vertex topology • $PV-SV = 202.91$ mm



PV/SV positions and flight line are reconstructed.
 Arrows show momentum directions as schematic rays; lengths are scaled.

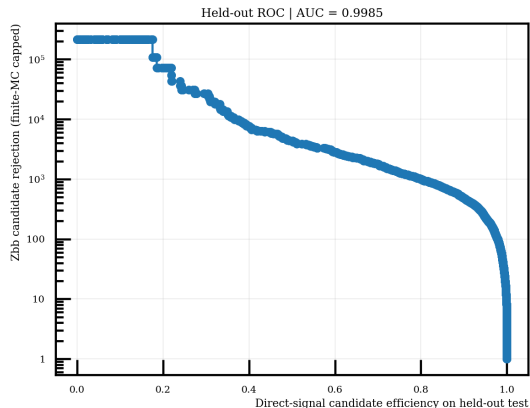
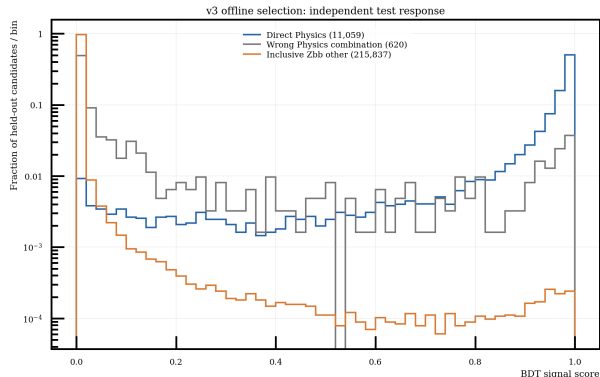
Momentum directions and isolation



Blue/green rings: exact ΔR boundaries around η/Λ^0 in $\eta-\phi$
 Solid/dashed: $\Delta R = 0.3, 0.5$ • grey ring: thrust-hemisphere boundary
 Grey/purple: other charged/neutral particles; crosses: opposite hemisphere
 Arrow lengths normalized; fitted daughter directions shown

PV and fitted Λ^0 SV in mm; ΔR boundaries are mapped from $\eta-\phi$ into direction space. One event for explanation, not an efficiency measurement.

Held-out response

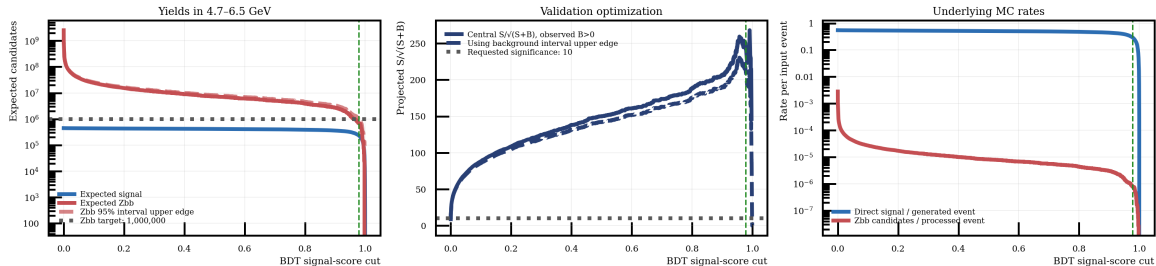


20 named reconstructed features; no mass, helicity angle, truth, event key, or charge sign in the fit.
Train/validation/test AUC: 0.99874 / 0.99864 / 0.99853. The tail rate still depends on observed background counts.

Model: docs/data/stage2_v3_bdt_1028/bdt_model.json; training summary in the same directory.

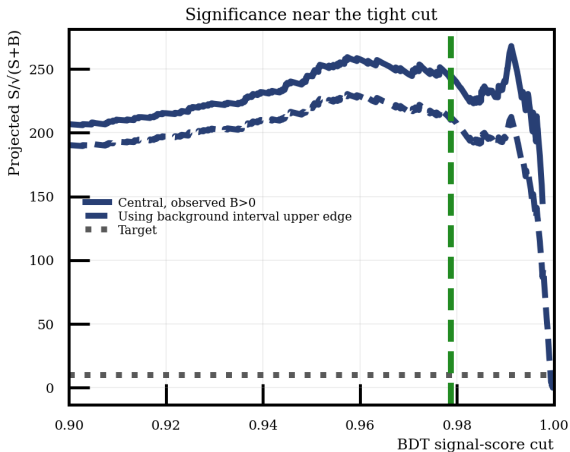
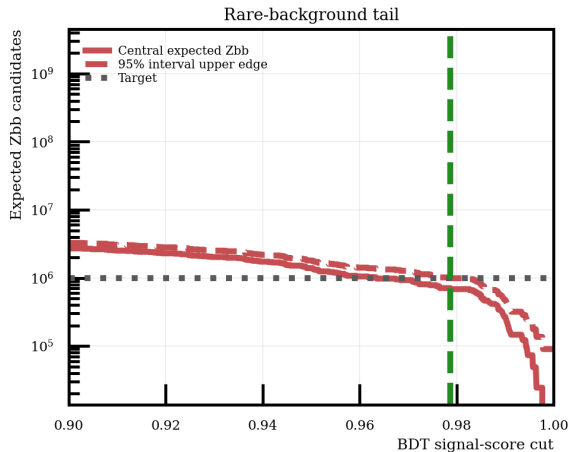
Validation score scan

Validation score scan | thresholds fixed before independent-test evaluation



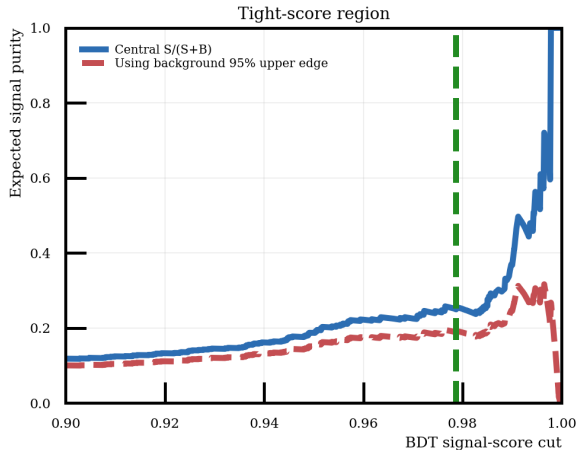
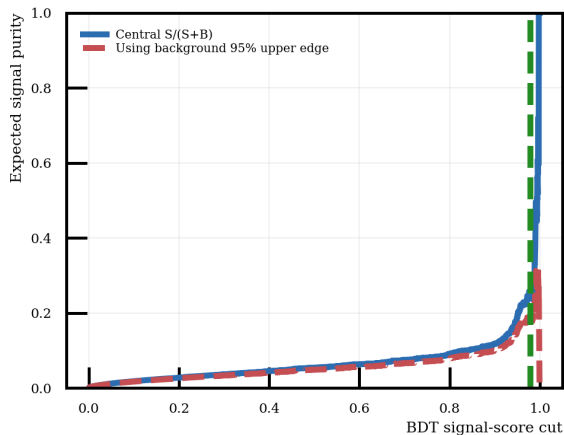
1,760 validation thresholds include dense near-one scores and exact background order statistics. Green: constrained inspection proposal at score ≥ 0.9787055 .

Tight tail: background count limits



Validation requires at least 20 observed Zbb candidates and an upper 95% expected B of at most one million for the marked proposal. A zero-MC tail is unresolved.

Purity does not yet reach the target

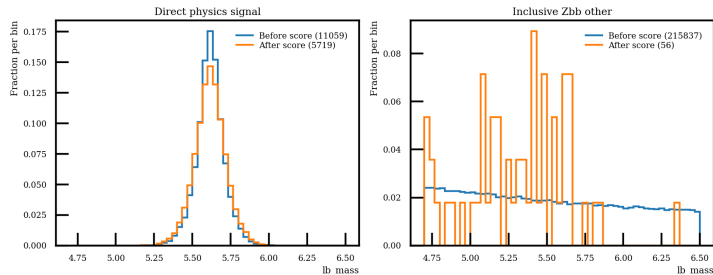


At score 0.995, only 3 validation Zbb candidates remain: central purity 54.8%, purity with 95% upper B 29.3%. At 0.999, zero observed B does not establish 100% purity.

Independent test at the validation-fixed proposal

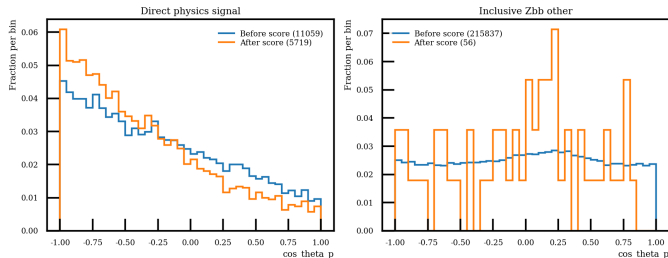
Score ≥ 0.9787055	Test result
Direct signal candidates	5,719
Zbb other candidates	56
Expected signal candidates	233,460
Expected Zbb candidates	657,999
Expected Zbb 95% interval	497,045–854,466
Central / upper-B purity	26.2% / 21.5%

Test denominators: 20,005 signal input events and 76,595,888 processed Zbb events. In a diagnostic 5.4–5.9 GeV interval, central purity is 43.7%; no fit window is adopted.



Mass and angle were excluded from training. Shape checks are required before a mass–angle fit.

Angular acceptance and next decision



- The score changes the direct-signal $\cos \theta_p$ shape through correlated reconstructed inputs.
- The marked score meets the background ceiling, but central purity is only 26.2% in the full mass interval.
- Tighter scores have 11, 6, 3, then 0 validation Zbb survivors in the sampled examples; more background statistics are needed for a high-purity claim.
- Refresh the remaining Zbb jobs, then process and normalize Z_{cc}/Z_{ss} before any mixed-background training.

PI choice pending: specify a purity requirement after reviewing plots. No BDT cut is adopted as the reference selection here.